

SEMINAR III • WEEK 2

Academic Problem-Solving

Teacher presentation — five days, built for modeling and discussion, not for reading off the slide.

DAY 1

How Do We Solve Problems?

Introduce the five-question routine · model one problem · students try it themselves

WARM-UP

When You Don't Know What to Do

"What could you do, even if you don't know the answer?"

Cue: put up a problem that looks unfamiliar or intimidating first. The point of this slide isn't the answer — it's proving students already have moves to make before they know the solution.

THE ROUTINE

The Five Questions

One routine, used all year, for math and non-math problems alike.

Introduce one question at a time — the next five slides.

QUESTION 1

What information do I have?

numbers

facts

definitions

labels

diagrams

graphs

tables

instructions

context

Say: don't calculate yet — first figure out what's actually available.

QUESTION 2

What am I trying to determine?

| *"I need to find _____."*

Have students say the sentence out loud before they touch a calculation — it stops blind calculating without a goal.

QUESTION 3

What skill or information could help?

addition

percentages

fractions

ratios

an equation

a graph

a formula

a definition

Students don't need the whole solution yet — just a lead worth following.

QUESTION 4

What strategy can I try?

write an equation

draw a diagram

make a table

estimate

eliminate choices

work backward

try a simpler example

The important word is **try**. It doesn't need to be guaranteed to work.

QUESTION 5

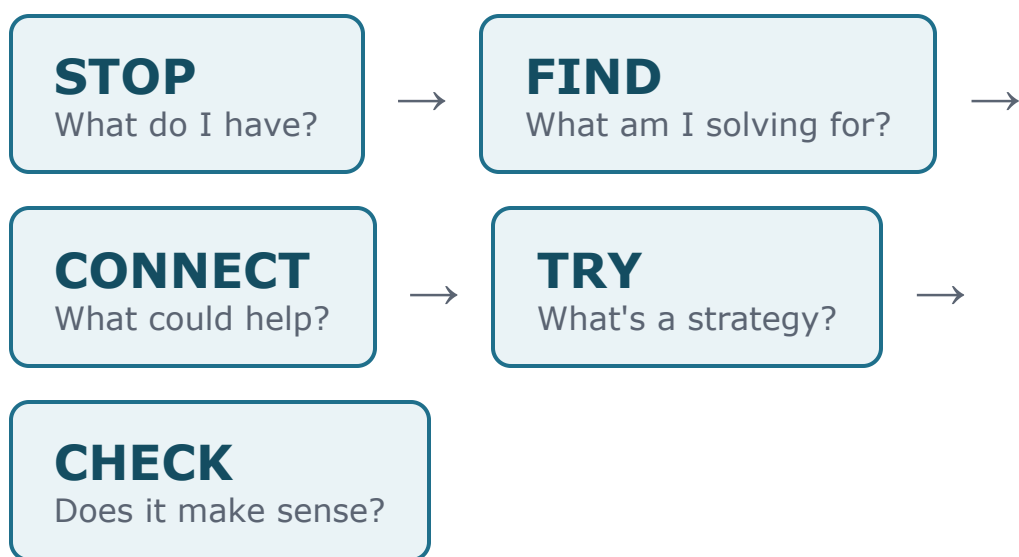
Does my answer make sense?

- Too large? Too small?
- Positive or negative?
- Does the unit make sense?
- Did I answer what was actually asked?

This question matters more and more as the quarter goes on — don't rush past it.

MEMORY AID

The Shortcut



The five questions stay the course's real language — this is just the fast version to reference mid-problem.

TEACHER THINK-ALoud

"Watch me think through one, out loud."

A student earns \$15 per hour and works 6 hours. How much before taxes?

Model it live, narrating all five questions as you go — STOP (\$15/hr, 6 hrs) → FIND (total earned) → CONNECT (rate × hours) → TRY (15×6) → CHECK (should land between \$75 and \$100 — \$90 does).

BEYOND MATH

A Problem Is Not Always a Math Problem

A project is due Friday. It needs a written response, a visual, and a presentation. It's Wednesday.

Run the same five questions on this — the routine isn't just for math class.

EXIT TICKET — DAY 1

Which of the five questions do you think you most often skip?

DAY 2

Why Did I Get It Wrong?

Introduce the five error types · analyze sample wrong answers · students classify 8–10 scenarios

SET-UP

"Wrong" Doesn't Tell Us Enough

Three students answer the same problem incorrectly — for three different reasons.

- **Student A** — doesn't know the rule
- **Student B** — knows the rule, calculates wrong
- **Student C** — solves correctly, answers a different question

"Do these students need the same help?"

THE FRAMEWORK

The Five Error Types

An incorrect answer doesn't automatically mean "I don't know how to do this."

ERROR TYPE 1

Knowledge Error

"There's something I need to know that I don't currently know or remember."

Example: doesn't remember how negative numbers behave during multiplication. — **Next step:** review the concept, study a worked example.

ERROR TYPE 2

Process Error

"I had the right general idea, but something went wrong in my steps."

Example: uses the correct formula but substitutes values into the wrong places. — **Next step:** compare with a correct example, write out intermediate steps.

ERROR TYPE 3

Execution Error

"I knew how to solve it, but I made a mistake while doing the work."

Example: copying 36 as 63; dropping a negative sign. — **Next step:** slow down, check work against an estimate.

ERROR TYPE 4

Comprehension Error

"I didn't correctly understand what the question was asking or telling me."

Example: finds the radius when asked for diameter; overlooks the word "NOT." — **Next step:** reread, underline the actual task.

ERROR TYPE 5

Strategy Error

"I understood the problem, but the approach I chose wasn't the best way to solve it."

Example: solving algebraically for several minutes when substituting answer choices would be faster. — **Next step:** ask "is there an easier way?"

RECAP

Error Types at a Glance

Type	Main Question
Knowledge	Did I know the information or skill I needed?
Process	Did I correctly follow the steps?
Execution	Did I make a smaller mistake doing something I knew how to do?
Comprehension	Did I understand the question correctly?
Strategy	Did I choose an effective approach?

Errors can overlap — the goal isn't a perfect label, it's answering "what should I do differently next time?"

PRACTICE

Error Detective

Let's classify these together. Click each answer choice to check yours.

ERROR DETECTIVE · 1 OF 5

$$6 + 3 \times 4 = ?$$

Student answers **36** ($6+3=9$, then $9 \times 4=36$)

Process Error

Correct calculations, wrong order of operations. Review order of operations and retry.

ERROR DETECTIVE · 2 OF 5

A \$80 jacket is discounted 25%.
Student correctly finds the discount is \$20
and answers **\$20**.

Comprehension Error

Calculated the discount, not the final price. Identify exactly what's being asked before answering.

ERROR DETECTIVE · 3 OF 5

$$-7 + 12 = ?$$

Student: "I don't remember what I'm supposed to do when one number is negative."

Knowledge Error

Review adding positive and negative numbers.

ERROR DETECTIVE · 4 OF 5

$$48 \div 6 = ?$$

Student correctly picks division, writes **$48 \div 6 = 6$** .

Execution Error

Right operation, wrong arithmetic. Check by multiplying the answer back out.

ERROR DETECTIVE · 5 OF 5

A multiple-choice problem could be solved fast by testing the four choices. Student instead builds a long multi-step equation, spends six minutes, doesn't finish. Reasoning is valid throughout.

Strategy Error

Look for a faster method before starting a long calculation.

EXIT TICKET — DAY 2

What type of error most likely occurred here, and what should the student do next?

Give one incorrect response as the prompt.

DAY 3

ACT Math Baseline

20–24 questions · diagnostic, not a grade

BEFORE STUDENTS BEGIN

Remind Them

- This is diagnostic — the goal is a starting point, not a score to defend.
- Attempting unfamiliar questions still gives useful information.
- Don't randomly click just to finish.

IF TIME REMAINS

Initial Reaction

- What felt familiar?
- What felt unfamiliar?
- Where did you feel most confident?

DAY 4

What Does My Baseline Tell Me?

Same score, different needs · students read their
own results

MINI-LESSON

Two students can get the same score and need completely different support.

Demonstrate with a quick example before students look at their own results.

READING YOUR RESULTS

Not Just a Score

Current Strengths

skills shown consistently

Developing Skills

partially shown / partly
remembered

Priority Skills — need real instruction or review

Avoid presenting results as just "you scored 9/24."

WHILE STUDENTS WORK

Help students tell these apart:

"I don't know this."

"I knew this but made a mistake."

DAY 5

Build Your Starting Strategy

Week 2 reflection · study hall / catch-up for
remaining time

WEEK 2 REFLECTION

Name Four Things

1. **One Strength** — something you do reasonably well
2. **One Priority** — something you want to improve
3. **One Strategy** — something to try when stuck
4. **One Checking Habit** — something to do before submitting an answer

THIS ROUTINE CONTINUES ALL QUARTER

Keep Using Both

STOP

→ FIND → CONNECT → TRY → CHECK

Knowledge

→ Process → Execution → Comprehension → Strategy

Week 3 begins Numbers & Operations — the first full run of this cycle on real content.